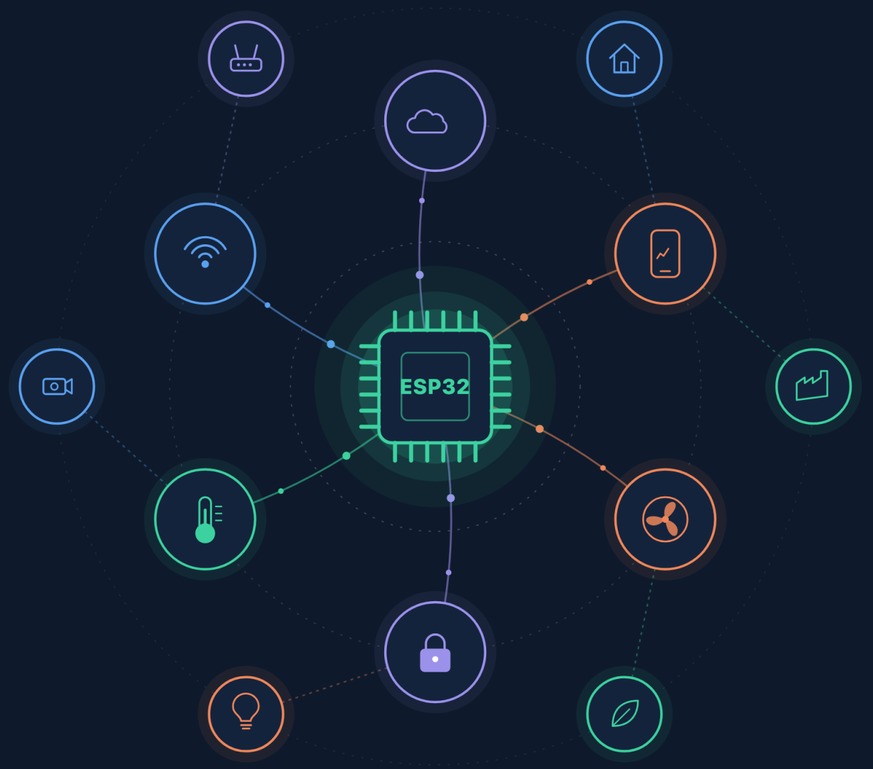
MODULE 2 · 45 MINUTES

How an IoT System Is Built

The five layers inside every IoT system, from a tower site to a data centre.



Mehdi Lotfinejad

Introduction to the Internet of Things

THE ROAD AHEAD

Eight modules, one Smart Site Monitor

PART 1 — DEVICES AND DATA

01
What is IoT?
Lecture

02
How it's built
You are here

03
Sensors and actuators
Lab 2

04
Getting online
Lab 3

PART 2 — CLOUD, AUTOMATION AND SECURITY

05
Cloud and dashboards
Lab 4

06
Automation
Lab 5

07
Security
Group activity

08
Real world + pitch
Lab 7

BY THE END OF THIS MODULE

You will be able to...

01 **Name** the five layers of an IoT system and what each one does

02 **Map** the five layers onto a tower site and a telco data centre

03 **Explain** the difference between a sensor and an actuator

04 **Compare** a microcontroller with a microprocessor

05 **Weigh** build, buy or telco-managed, and selling it as a product

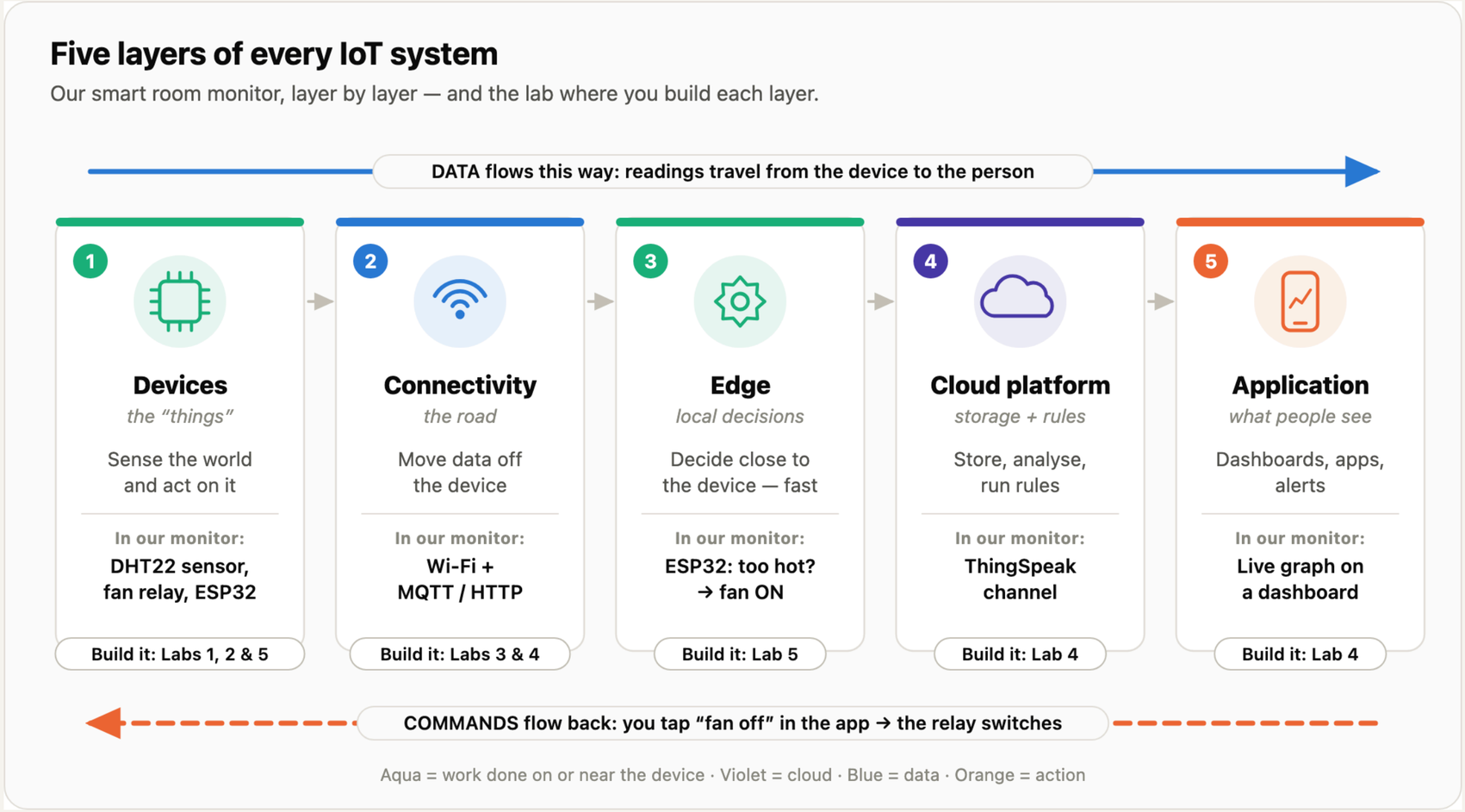
Part A

The five layers

One Smart Site Monitor, broken down layer by layer.

THE BIG PICTURE

Five layers of every IoT system



LAYER 1 · DEVICES

Three ingredients in every device

Sensors

Measure something, like shelter temperature or fuel level.

Actuators

Do something, like switching on extra cooling.

A small computer

Usually a microcontroller that runs the program.

In the labs, your device is a simulated **ESP32** board.

LAYER 2 · CONNECTIVITY

The road the data travels on



The link

How the signal travels: a mobile network, a cable or Wi-Fi.



The language

What the device speaks, like MQTT or HTTP.



A gateway

A middleman box that serves many nearby devices.

At a tower site, the site’s monitoring box is the gateway. At home, your Wi-Fi router is one.

LAYER 3 · EDGE COMPUTING

Edge computing means doing some of the thinking close to where the data is collected.

On the device itself, or on a gateway nearby — not far away in the cloud. It is faster, and it keeps working when the connection doesn’t.

LAYER 4 · CLOUD PLATFORM

What a cloud platform does

Stores

Keeps old readings, so you see history.

Processes

Works things out, like a daily average.

Runs rules

“Above 28 °C? Send me a message.”

Manages devices

Knows which devices exist and may connect.

In Lab 4 you use a free cloud platform, **ThingSpeak**. Module 5 covers this layer.



Photo: Daoducquan, CC BY-SA 4.0, via Wikimedia Commons

LAYER 5 · APPLICATION

What people see and use

- **Dashboards** in the NOC, with live graphs
- **Alerts** by app, SMS or email
- **Reports** for managers
- **Work orders** that send a field engineer
- **Other systems**, like billing

PAUSE AND CHECK

Our Smart Site Monitor switches on extra cooling in the shelter by itself, without asking the cloud. Which layer is making that decision?

Devices · Connectivity · Edge · Cloud platform · Application

ONE MAP, TWO CASES

The five layers at a tower and a data centre

Layer	Tower site	Telco data centre
1 Devices	Shelter heat, fuel, battery, door	Rack heat, humidity, leak, door
2 Connectivity	Cellular link (mobile network)	Cable or Wi-Fi
3 Edge	Site monitoring box (RMS)	Gateway, BMS or DCIM on site
4 Platform	Monitoring platform in the NOC	Monitoring platform
5 Application	NOC alarms, field engineer sent	Operations dashboard and alerts

RMS remote monitoring system · **NOC** network operations centre · **BMS** building management system · **DCIM** data-centre infrastructure management

Part B

Inside the device

Senses, muscles and a brain.

THE DEVICE’S SENSES AND MUSCLES

Sensors and actuators

	Sensor (input)	Actuator (output)
Job	Measures the world	Changes the world
Like your…	Senses	Muscles
Examples	Temperature, fuel-level, door and smoke sensors	LED, buzzer, motor, relay (an electric switch)
In our monitor	Shelter temperature and humidity sensor	Relay that switches extra cooling

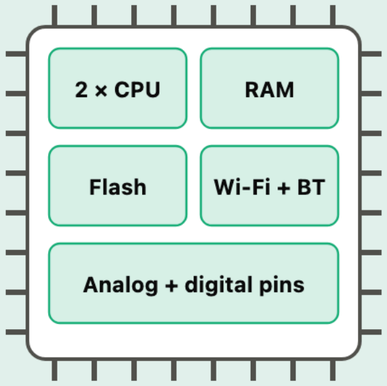
Ask: **which way is the information going?** In from the world = sensor. Out into the world = actuator.

One chip, one job vs a small computer

One chip, one job vs a small computer

A microcontroller packs everything onto one chip. A microprocessor needs a board of helpers and an OS.

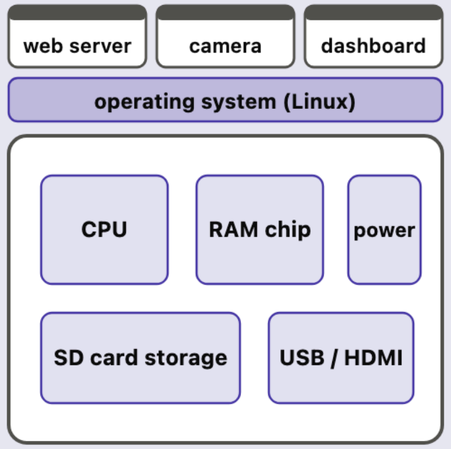
MICROCONTROLLER (e.g. ESP32)



everything on ONE chip

- **Runs ONE program,** from the moment power is on
- **Starts instantly** no booting
- **Deep sleep ≈ 10 µA** months–years on batteries
- **Low cost** cheap enough to put in anything

MICROPROCESSOR BOARD (e.g. Raspberry Pi)



several chips on a board

- **Runs an OS +** many programs at once
- **Boots in seconds** like a small PC
- **Draws watts** usually mains-powered
- **Great for video, AI** and heavy software

SIDE BY SIDE

Microcontroller or microprocessor?

	Microcontroller (MCU)	Microprocessor (MPU)
Like a…	Kitchen timer: one job, done reliably	Laptop: many jobs, more looking after
Runs	One program, straight away	An operating system and many programs
Power use	Very low: months or years on batteries	Higher: usually mains power
Start-up	Almost instant	Seconds to a minute (it boots)
Cost	A few US dollars	Tens of US dollars and up
Good at	Reading sensors, switching things	Video, heavy data, a local dashboard
Examples	ESP32, Arduino boards, STM32	Raspberry Pi, the chip in your phone

Reads sensors and flips switches? **Microcontroller.** Video or lots of software? **Microprocessor.**

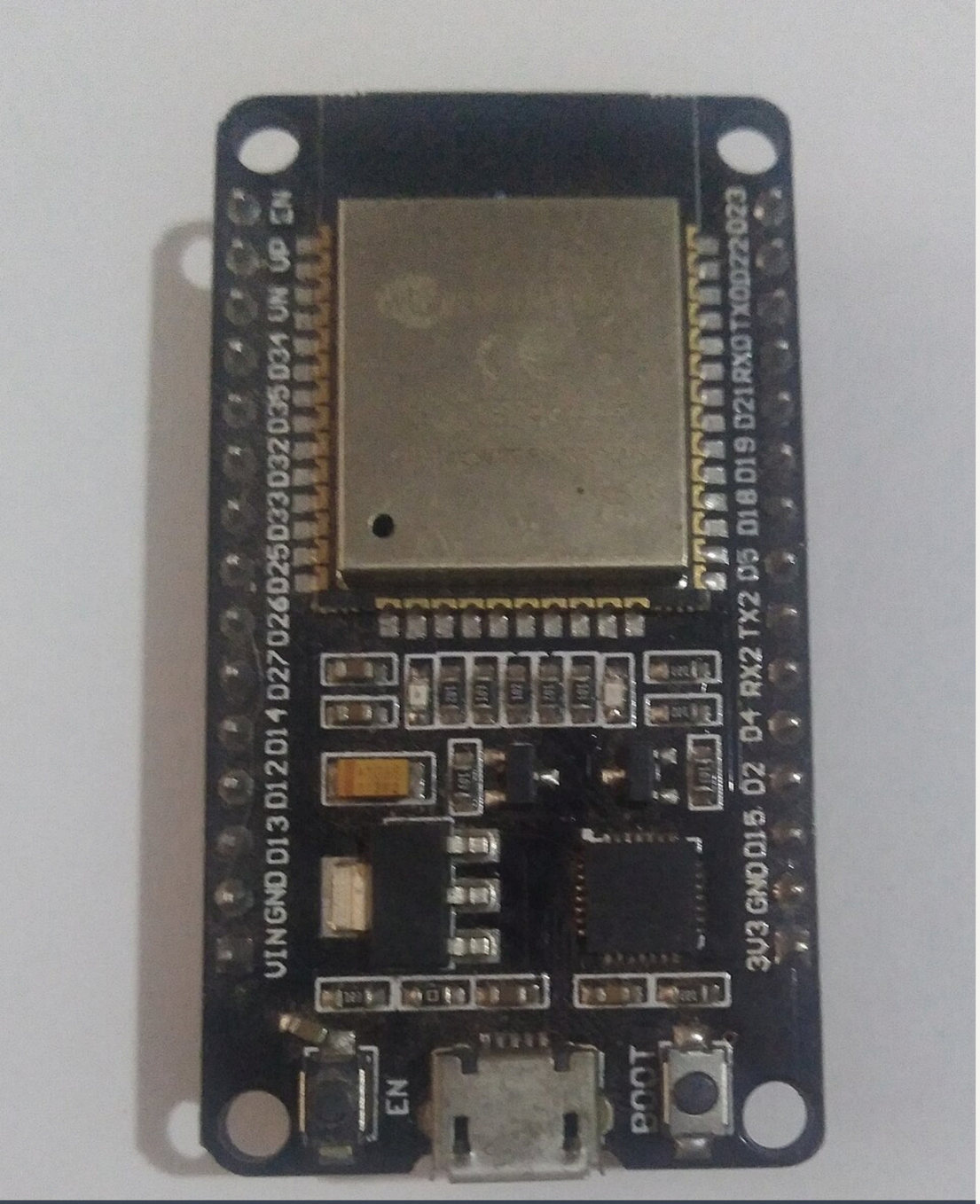


Photo: Edwiyanto, CC BY-SA 4.0, via Wikimedia Commons

THE DEVICE YOU'LL USE

Meet the ESP32

- The brain of our **Smart Site Monitor**
- **Wi-Fi and Bluetooth** built into the chip
- **Cheap**, and uses little power when resting
- **Lots of pins** for sensors and actuators

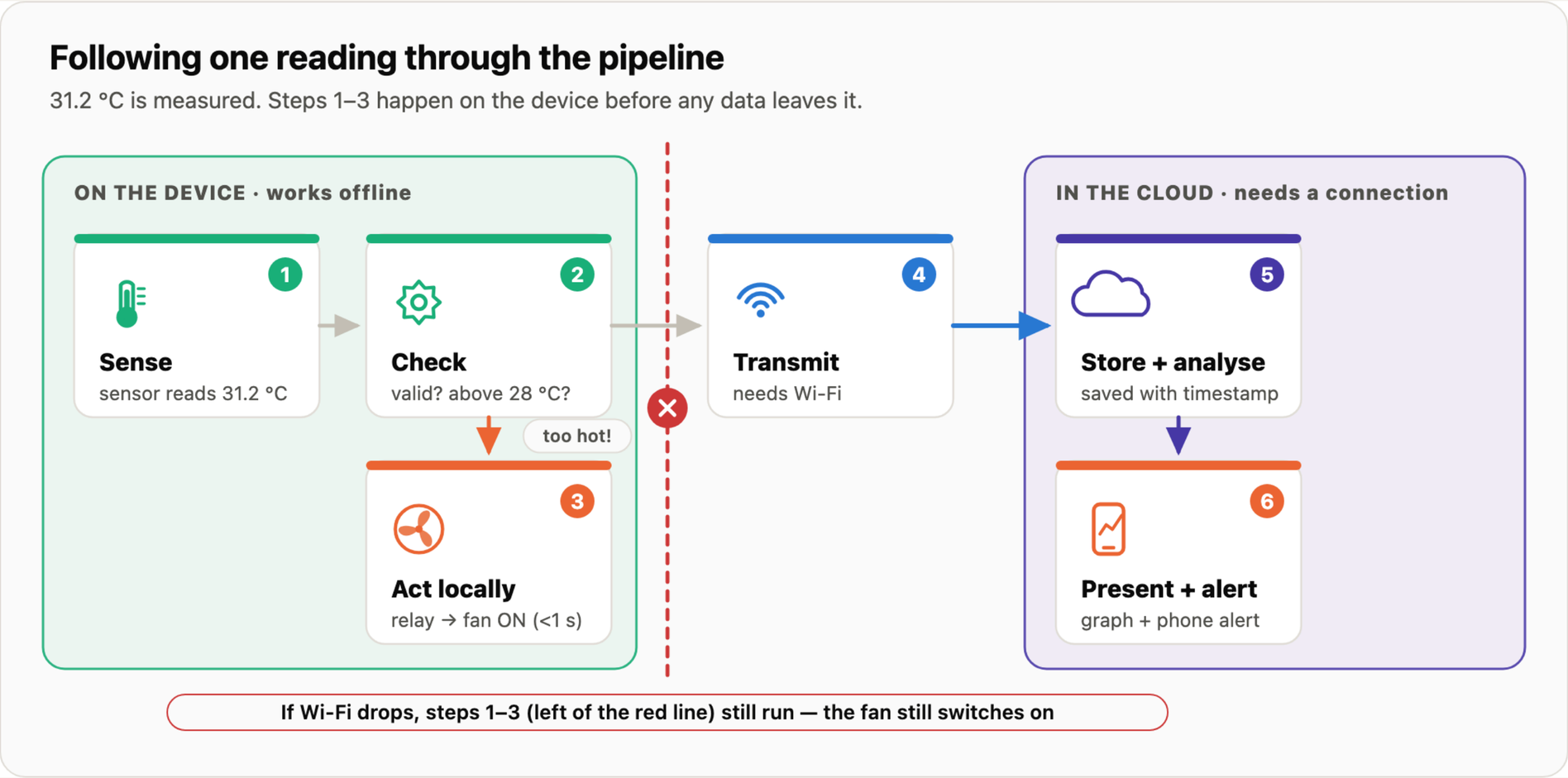
Part C

One reading, and who builds it

From a sensor, through the layers, to a person — and three ways to get there.

THE DATA PIPELINE

One reading’s journey



PAUSE AND CHECK

In the pipeline, which steps would still work if the network link went down?

1 Sense · 2 Check · 3 Act locally · 4 Transmit · 5 Store · 6 Present. Talk to your neighbour for one minute.

BUSINESS LENS · WHO BUILDS IT?

Build, buy or telco-managed



Build in-house

Cost: parts and staff time. **Speed:** slowest. **Control:** full.



Buy from a vendor

Cost: price plus licences. **Speed:** fast. **Control:** the vendor’s.



Telco-managed

Cost: a monthly fee. **Speed:** fast. **For a telco:** also a product to sell.

Telcos don’t only use IoT, they sell it: **Maxis** sells NB-IoT, 4G M2M and 5G IoT plans (Maxis Business website).

COMMON MISCONCEPTIONS

Three ideas that trip people up

✗ **Myth** *An IoT device must connect directly to the internet.*	✓ **Fact** Many talk only to a nearby gateway, which reaches the internet.
✗ **Myth** *The cloud does all the thinking.*	✓ **Fact** More and more decisions happen at the edge: faster, and offline too.
✗ **Myth** *A Raspberry Pi and an Arduino are the same thing.*	✓ **Fact** Different families: an OS on a microprocessor vs one program.

TEAM ACTIVITY · 5 MIN

Map a tower site or a telco data centre

1. Default: a tower site. Or choose a telco data centre.
2. Draw five boxes: devices, connectivity, edge, cloud, application.
3. Put one real part of the site into each box.
4. Which decision must happen at the edge? Why?
5. Decision owner: build, buy or telco-managed?

IN TELCO OPERATIONS

Smart tower monitoring, day to day

	Today	With IoT	The benefit
NOC alarm handling	Site off air; the NOC can’t see why	Sensors show why: power, heat, door	Faster fault repair
Field visits	Drive out, just to look	Check the site on the NOC screen	Fewer site visits
Site energy	Levels checked only on visits	Readings arrive in the NOC all day	Less diesel, less theft

Same five layers: sensors at the site, a cellular link, the RMS box, the NOC platform and alerts.

KEY TAKEAWAYS

What to remember

- Five layers: **devices, connectivity, edge, cloud, application**
- The same layers fit a **tower site** and a **data centre**
- Sensors bring information **in**; actuators push action **out**
- **Microcontrollers** do one job cheaply; **microprocessors** run an OS
- Urgent decisions stay at the **edge**, even when the link drops
- **Build, buy or telco-managed**; a telco can also **sell** it

CHECK YOUR UNDERSTANDING

Quick quiz

Q1 List the five layers, in order from the device to the person.

Q2 At a tower site, is the shelter door contact a sensor or an actuator? And the relay that starts the generator?

Q3 A fuel sensor at a remote tower must run for years on a battery. Which brain: a microcontroller or a microprocessor?

Q4 A customer asks our telco to run their IoT. Name one strength and one risk.

UP NEXT · MODULE 3

Sensors, Actuators, and Reading Data

We zoom in on Layer 1: the sensors at a tower site and in a data centre. Then your first lab.

Mehdi Lotfinejad

Book, labs and slides: github.com/datatweets/intro-to-iot-workshop